

00 — Platform Basis: Scope, Design Point, Shared Physics & Doctrines

v1.5 — foundation document for both tracks

Function: Everything both tracks share: deployment scope, the governing design point, the shared psychrometric argument, the generalized airflow-moisture model, the ventilation/CO₂ stack, the exhaust-vapor recovery doctrine, and the safety-critical requirements register. Track documents (10–12 liquid, 20–22 solid, 30 integration) reference this document rather than restating it.

New to this lineage? §9 carries the notation, the symbol table, and the key to the F / X / P / test identifiers used throughout every document.

1. Scope — land and marine

The system targets **both land and marine deployment** in humid-tropical (and by extension any humid) climates. Design rules are written to the *harshes common case* so one document set covers both:

Aspect	Marine (governing case)	Land relaxations
Raw-water sink	Seawater ~29 °C (tropical surface)	Lake/river/well water, often cooler (22–28 °C tropics; lower elsewhere) → every sorbent floor improves (see §3 leverage). No water sink: evaporative fluid cooler (approaches wet bulb ~30 °C — near parity) or dry cooler (approaches dry bulb 32 °C — floors rise ~1.5–2 g/kg; state the penalty)
Motion	Heel/roll to ~15–20°; flooded-mode fallbacks, staged headers, anti-slosh rules	Stationary: motion provisions optional, retained as shipped hardware margin
Salt aerosol at intake	Mandatory intake filtration / sealed paths	Coastal: keep; inland: may relax intake filtering only
Chloride materials rules	Full two-worlds discipline (doc 11 §1)	Never relaxed — the desiccant itself is the chloride source
Envelope	~100 m ³ occupied reference (yacht interior / small dwelling / cabin / shelter)	Same physics; volume thresholds per §6 ladder

Terminology: "platform" = vessel or land site; "raw-water sink" = seawater or its land equivalent. Marine values are used in all reference numbers; land is margin.

2. Governing design point (sole)

DP-A: 32 °C / 80% RH · ω = 24.2 g/kg · dew point 28.1 °C · raw-water sink ~29 °C. DP-A is the **continuous-duty maximum rating for every mode**. Milder ambients are operating margin (every duty term shrinks monotonically with ambient ω); rare excursions above DP-A are handled transiently — moisture-battery draw-down, setpoint relaxation, boost — never by sizing. A prior secondary point (30 °C / 75%) is retired; numbers derived from it in archived documents are historical records.

Saturation humidity ratios used throughout (Magnus, 101.325 kPa): 26 °C → 21.3, 27 °C → 22.6, 28 °C → 24.1, 29 °C → 25.6, 30 °C → 27.1 g/kg.

Air-density basis: volumetric-to-mass air conversions throughout the lineage use **p = 1.2 kg/m³** (standard air), not the ~1.15 kg/m³ of moist air at DP-A itself. Every mass flow — and every duty derived from one — therefore runs ~5% high. The direction is deliberate: it oversizes rather than undersizes. Stated here so it is an assumption on the record rather than one implied by the arithmetic.

3. The shared physics — why a desiccant is non-negotiable

Comfort in humid heat requires removing water from air, and only two things do that: a surface below the dew point (refrigeration) or a sorbent whose surface vapor pressure sits below ambient (a desiccant). At DP-A every ambient sink — raw water ~29 °C, outside air — sits **at or above the 28 °C dew point**, so:

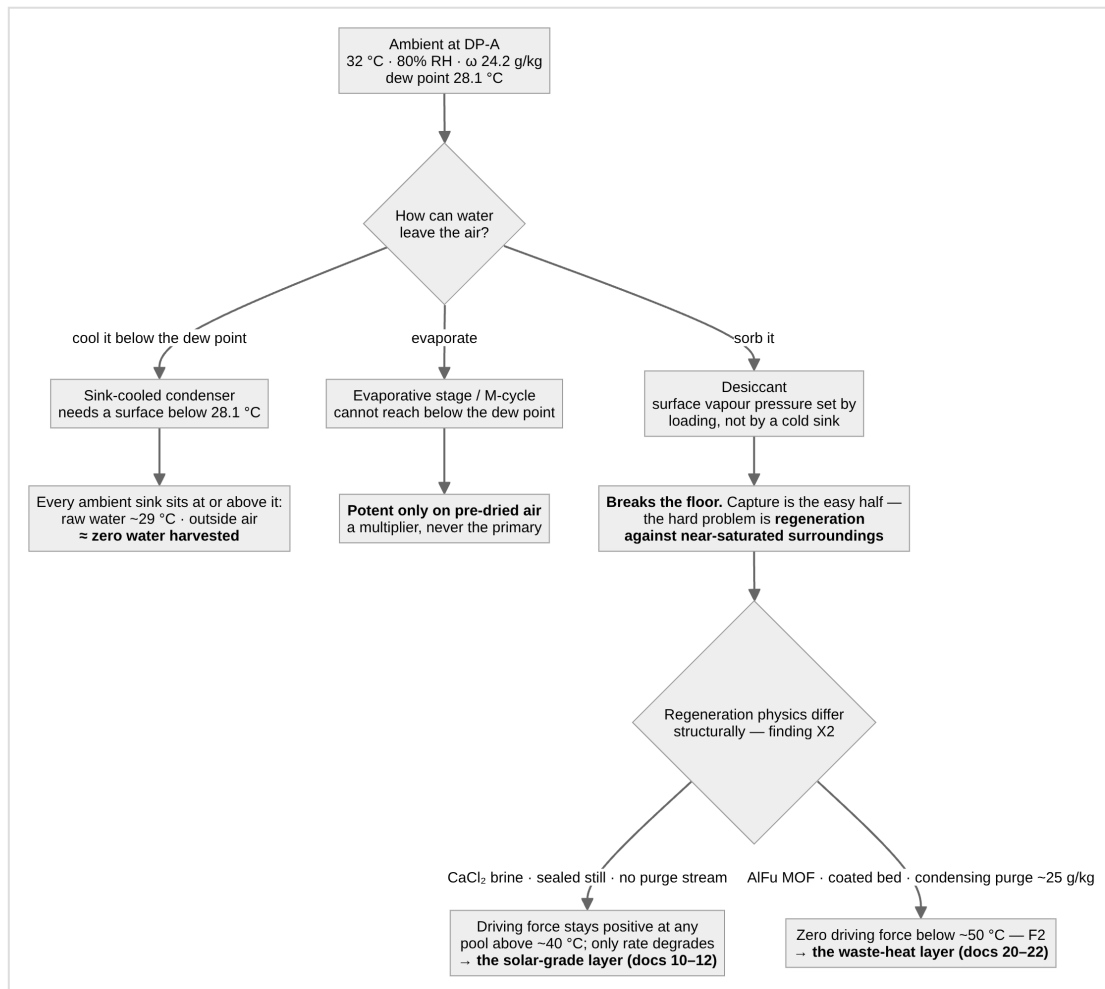
- a sink-cooled condenser harvests ≈ zero water from ambient air;
- evaporative cooling (including the M-cycle) cannot go below the dew point — potent only once air is *pre-dried*;
- the desiccant breaks the floor because its surface vapor pressure is set by loading/concentration, not by a cold sink. The hard problem is **regeneration against near-saturated surroundings**, never capture.

Terminology (equivalence). The *dew-point indirect evaporative stage* — short form *dew-point IEC* — is the cycle published as the **Maisotsenko cycle** and written **M-cycle** throughout this lineage and in the HVAC literature. All four terms denote the same stage. The

generic forms are recorded so the disclosure does not depend on an eponym or a trademark; the eponymous forms are retained in the text so the record stays discoverable under the terms an examiner or searcher will actually use.

Two sorbent families implement this: a CaCl_2 brine (liquid track, docs 10–12) and an AlFu MOF coated exchanger (solid track, docs 20–22). Their regeneration physics differ structurally — **finding X2**: the liquid track's sealed still has no purge stream, so its driving force (hot-pool Pv vs ~4–5 kPa sink-cooled condenser) stays positive at any pool temperature above ~40 °C and only *rate* degrades at low grade; the solid bed desorbs into a condensing purge (~25 g/kg) and hits a zero-driving-force wall below ~50 °C (F2). Consequence: **the liquid track is the solar-grade layer; the solid track is the high-duty waste-heat layer.**

Figure 3.1 — the dew-point floor, and the one route through it. No number in this figure is new; it restates the argument above so it can be checked at a glance.



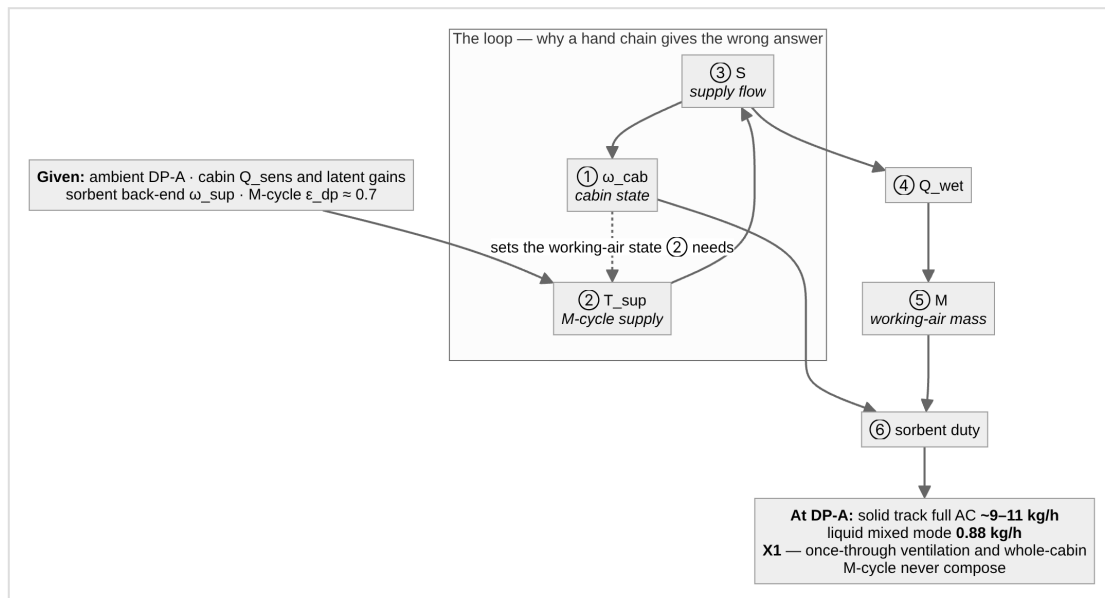
Raw-water leverage (both tracks): cooler sink water deepens every floor — ~0.5–0.7 g/kg per °C on the brine side; land installs with cool groundwater get this for free.

4. The generalized airflow-moisture model

Closed steady-state balance replacing hand chains; the sorbent back-end enters only through the supply humidity ω_{sup} (AlFu ~8 g/kg; brine per the doc 12 floor matrix). Unknowns solved simultaneously:

1. $\omega_{\text{cab}} = \omega_{\text{sup}} + \text{latent_gains} / S$ (cabin steady state)
2. $T_{\text{sup}} = T_{\text{in}} - \epsilon_{\text{dp}} \cdot (T_{\text{in}} - T_{\text{dp}}(\omega_{\text{work}}))$ (M-cycle, $\epsilon_{\text{dp}} \approx 0.7$)
3. $S = Q_{\text{sens}} / (c_p \cdot (T_{\text{cab}} - T_{\text{sup}}))$
4. $Q_{\text{wet}} = S \cdot c_p \cdot (T_{\text{in}} - T_{\text{sup}})$
5. $M = Q_{\text{wet}} / \Delta h_{\text{work}}$, $\Delta h_{\text{work}} = h_{\text{sat}}(T_{\text{exh}}) - h(\text{working-air entry state})$
6. Sorbent duty = $(S - M)(\omega_{\text{cab}} - \omega_{\text{sup}}) + M(\omega_{\text{amb}} - \omega_{\text{sup}})$

Figure 4.1 — the six equations close on each other. The arrows are dependencies, not a running order: S appears in ① and is defined by ③, which needs T_{sup} from ②, which needs the working-air state that ① sets. That circularity is the reason a hand chain gives the wrong answer and a simultaneous solve is required.



Load-bearing outputs at DP-A: solid track full-AC duty **~9–11 kg/h** (doc 20 §5); liquid mixed-mode duty **0.88 kg/h** (doc 12 §1); and **finding X1** — once-through ventilation and whole-cabin M-cycle cooling never compose (drying 370–750 m³/h of ambient supply costs 5.5–11 kg/h). A full transient model (task T2) supersedes this steady state; it does not replace it.

5. Ventilation & CO₂ — the four-layer stack (X6/X7/X9/X10, spec P17)

Specification (safety-critical): cabin CO₂ <1,000 ppm at all times, in every mode including sealed; alarm + forced boost at 2,000 ppm. Four occupants generate ~3.4 kg CO₂/day (**crew total**, not per person: 0.072 m³/h awake / 0.047 asleep); CO₂ leaves only with air exchanged, so recirculation, cascades, and ERVs move moisture, never CO₂.

- **Layer 0 — eliminate (X9): all-electric galley, no gas or combustion appliances aboard/on-site.** One 2 kW burner emits ~0.26 m³/h CO₂ (3.6× the crew) plus ~0.25 kg/h combustion water; induction deletes both (~2 kWh_e/day).
- **Layer 1 — ventilate efficiently.** 48 m³/h CO₂-governed fresh-air floor (12 m³/h·person; also carries bioeffluents, odors, O₂). **ε_{lat} ≥ 0.8 counterflow ERV** on the fresh/exhaust pair (CO₂ crossover <5%, test E); demand-controlled ventilation; **displacement ducting** — supply low to berths first, exhaust pickups high in every closable room (closed doors otherwise run +2,000–2,900 ppm over the main space); **per-room NDIR sensors, interlock keyed to the maximum reading.** Marginal cost of fresh air at DP-A: (1–ε_{lat}) × 10.4 g/kg ≈ 0.06 kWh_{heat}/day per m³/h at ε 0.8 (**ERV'd basis** — the bare-fresh equivalent is ~5× higher, and the two are quoted separately below).
- **Layer 2 — scrub the increment (X10): the CO₂ battery, required-PENDING test J.** Two-bed solid-amine thermal-swing sorbent (Lewatit VP OC 1065-class, or DIY PEI-on-silica) in the recirculation branch, post-dehumidification (45–55% RH is near-optimal for amine chemistry). Duty at the ventilation floor: ~1.6 kg/day; two ~3 kg beds on ~90-min half-cycles; ~91 m³/h at 50% single-pass capture rides the existing recirc flow (+50–150 Pa); **regeneration 85–95 °C off the same heat bus, ~1.0–1.3 kWh/kg → ~1.6–2.5 kWh/day**; CO₂-rich purge vented out (§7 rule 1). Loaded beds store the overnight obligation (~0.3 kg) → regeneration is solar-window schedulable. **Bed chemistry follows the heat grade (X11):** the scrubbing duty is chemistry-agnostic; select the sorbent by the highest heat grade reliably available, top down. **≥ ~130 °C tap (waste-heat platforms): K₂CO₃-on-apolar-carbon TSA** — same bed envelope (~4–5 kg/bed at ~0.5–0.65 mmol/g working), carbonation *wants* the 45–55% RH placement (water is a reagent), regeneration 130–150 °C, food-grade chemistry with no amine-slip or oxidative-fade pathway (the breathing-air gate becomes an alkaline-carryover assay), required-PENDING test J-K; **alumina and MgO supports prohibited** (irreversible double-salt deactivation). **Solar grade only (60–95 °C): the amine resin above** — potash is equilibrium-dead below ~120 °C (an F2-analogue wall; the ETC's 93 °C ceiling cannot regenerate it). Combinations run the ladder for redundancy: a platform fitting potash as primary retains a small amine bed (or accepts the open-mode fallback) for sealed operation through a heat outage. **This is the only path that holds <1,000 ppm sealed.** Fallback if test J fails: oversized ERV + DCV at 120–190 m³/h → ~800–1,000 ppm, open conditions only, no sealed-mode guarantee. Rejected alternatives (quantified in doc 12 §5): zeolites (water outcompetes CO₂), seawater absorption (>6 m³/h equilibrated + re-humidifies the air), membranes at 0.1 kPa, algae (150+ kWh/day of light), liquid amines (slip into breathing air), chemical scrubbing (~15 kg/day soda lime — **emergency consumable only**).
- **Layer 3 — interlock.** CO₂-governed fresh damper with a **mechanical minimum stop** (no controller fault or economy setting can close ventilation below the floor); recirculation-only operation prohibited while occupied; scrubbing never substitutes for the ventilation floor.

Control law: optimize marginal ventilation against marginal scrubbing on heat availability — heat-rich, open the damper toward ~600–800 ppm; heat-scarce or sealed, the bed holds <1,000 at the floor.

CO₂ dose-response at the floor and below (4 occupants; why the interlock exists): 48 m³/h → 1,920 ppm awake; 36 → 2,420; 24 → 3,420; 12 → 6,420; sealed (infiltration only) → ~7,600. Each 12 m³/h of fresh air cut saves ~3.8 kWh/day of **bare-fresh** heat at DP-A (≈0.32 kWh/day per m³/h un-recovered; with the ε 0.8 ERV fitted, the same cut saves only ~0.7 kWh/day) — a standing economic temptation the interlock forecloses.

6. Volume-threshold ladder at DP-A (liquid track; solid track is full-AC class)

Mode	Serviceable volume	Limit type
Once-through cascade M-cycle cooling	~2–5 m³ per outlet (74–112 W)	dew point of working air
Unattended, 1 contactor, holding 13 g/kg	~90 m³ (×2 per contactor)	contactor capacity vs 0.1 ACH
Unattended, 1 contactor, mold-safe (60–65% RH)	~160–215 m³	"
Once-through occupied (pressure integrity)	~110–240 m³	supply ≥ 2–3× infiltration
Mixed-mode occupied (baseline)	~300–500 m³ with a 3-cell bank	bank capacity + heat budget
Whole-cabin AC (recirc + hot-regen brine)	100 m³-class at ~10.6 kg/h, ~11 kW heat	deferred — converges with the solid track

Diagram: `diagrams/dpa-volume-thresholds.svg`.

7. Exhaust-vapor recovery doctrine (X8)

All *deliberately saturated* process exhausts terminate in distillation recovery when marginal regeneration heat is available; heat is recaptured where practical. Diagram: `diagrams/exhaust-recovery-doctrine.svg`.

- 1. **No double duty.** A stream is a rejection sink *or* a recovery source, never both — recapturing deliberately exported moisture re-imports the load just paid to reject.
- 2. **ERV protection.** The ducted room exhaust is reserved as the ERV moisture sink; the ERV outlet leaves near-saturated *by design* and is discharged un-recaptured.
- 3. **Saturated process exhausts end in recovery:** the M-cycle working stream closes on the desiccant loop (solid track: design intent, doc 20 §8; liquid track: valved free-heat option at ~1.3 kWh/L, doc 10 §6); the air-swept regeneration exhaust (degraded mode) gains a raw-water condenser — its condensate is **technical-grade only, PENDING TDS assay** (entrainment).
- 4. **Heat-recapture order:** DHW tank first (demand-capped ~1–3 kWh/day) → bed-to-bed / brine-to-brine recuperation → raw-water sink last. Adsorption heat releases only a few kelvin above the sink — reject it, don't chase it.

8. Safety-critical requirements register

Any build omitting these departs from this design:

- 1. CO₂ interlock per §5: <1,000 ppm target, 2,000 alarm/boost, per-room max sensing, mechanical minimum stop on the fresh damper.
- 2. No gas or combustion appliances in the conditioned envelope (X9).
- 3. Aerosol control chain + steel-coupon acceptance (test B) before any liquid contactor connects to breathing air (doc 11 §3).
- 4. CO₂-sorbent breathing-air assay before any bed connects to breathing air: amine/ammonia slip (test J) for amine beds; alkaline particulate/mist carryover (test J-K) for carbonate beds.
- 5. Distillate potability: independent water-quality test before regular consumption; air-swept-condenser water never enters potable or M-cycle-feed service before a TDS assay clears it.
- 6. Crystallization interlock: 42 wt% high-SG cutoff unless brine ≥22 °C guaranteed; 43 wt% storage cap (doc 11 §4).
- 7. Two-worlds materials rule on every brine/raw-water-wetted component (doc 11 §1).

9. Conventions, notation, and the ID key

Confidence grades on every quantitative claim: **measured** · **procurement-grade** · **sizing-grade** · **PENDING** <test/task>. DP-A on every number. Diagrams: Mermaid inside documents; the SVG architecture set lives in `diagrams/`. Psychrometric chains as state tables. Surgical edits with version bumps.

9.1 Identifier scheme

Every identifier in this lineage is **stable and never renumbered** — a reader returning to X8 years later must find the same finding. Gaps are deliberate and are part of the audit trail: numbers are assigned in the order items enter the queue, so a missing number means "drafted, not yet published", never "withdrawn".

Prefix	Meaning	Lives in
F1 – F6	Solid-track findings	doc 40 §1
X1 – X14	Cross-track findings — the ones that changed both tracks	doc 40 §2
P1 – P20	Patches from the archived predecessors, applied in place	throughout
P17	The CO ₂ specification (safety-critical)	doc 00 §5, §8
A – L	Liquid-track bench tests (A2, A3, J–K are variants)	doc 12 §4
T1 – T9	Solid-track sorbent/synthesis tasks	doc 21, doc 22
M1 – M4	Solid-track module measurements	doc 22 §6

Prefix	Meaning	Lives in
CHK-nnn	Consistency observations raised while re-deriving figures	register Checks sheet
LQ-, SD-, CO2-, IN-, UP-	Parameter-register row IDs (liquid, solid, CO ₂ , integration, upgrade)	parameter_register.xlsx

The procurement layer for every test above — what to buy, in what order, to what spec — is doc 41. It adds no experiment and decides nothing; where it and a track document disagree, the track document wins.

9.2 Symbols

Symbol	Quantity	Unit
ω	Humidity ratio (absolute humidity)	g water / kg dry air
Δq	Sorbent working capacity per swing	g water / g sorbent
a_w	Water activity of a brine — the equilibrium RH it holds	—
ϵ	Recovery effectiveness (ϵ_{lat} = latent)	—
NTU	Number of transfer units (contactor sizing)	—
ΔP	Pressure drop across a face or train	Pa
T_{dp}	Dew-point temperature	°C
GOR	Gained output ratio — distillate per unit heat	—
COP	Coefficient of performance	—
ERH	Equilibrium relative humidity over a brine	%

9.3 Abbreviations

Architecture. DP-A, the sole design point (§2) · **DCHX**, desiccant-coated heat exchanger (the solid track's module) · **LDAC**, liquid-desiccant air conditioning · **ERV**, enthalpy recovery ventilator · **DCV**, demand-controlled ventilation · **VC-AC**, vapour-compression air conditioning · **AHT**, absorption heat transformer (X12) · **MVR**, mechanical vapour recompression · **HDH**, humidification–dehumidification desalination · **RO**, reverse osmosis.

Cycles. **IEC**, indirect evaporative cooling; the *dew-point IEC* stage is the same cycle as the **M-cycle** / Maisotsenko cycle — see §3 · **TSA**, thermal swing adsorption (the CO₂ beds, X10/X11).

Materials. **AlFu**, aluminium fumarate (MIL-53(Al)-FA, Basolite A520) · **MOF**, metal–organic framework · **CaCl₂**, the liquid track's brine.

Heat sources. **ETC**, evacuated-tube collector · **PVT**, photovoltaic–thermal panel · **PTC**, positive-temperature-coefficient (resistive) heater.

Assays. **PXRD**, powder X-ray diffraction · **DVS**, dynamic vapour sorption · **TDS**, total dissolved solids.

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Version history

- **v1.0** — New lineage. Consolidates the shared basis from the archived core-01 §1, 01 §1–2, and 06 §§1–3, 9–11; land+marine scope made explicit; DP-A sole; CO₂ stack (P17, X9/X10), X8 doctrine, and safety register carried in place.
- **v1.1** — X11: heat-grade-dependent CO₂-bed ladder added to §5 Layer 2 (K₂CO₃/apolar-carbon TSA for ≥ ~130 °C taps, test J-K; alumina/MgO support prohibition; amine bed retained at solar grade and as outage fallback); safety-register item 4 generalized to cover the carbonate alkaline-carryover assay.
- **v1.2** — Clarifying pass from the parameter register (doc 50 §3.5), no design figure changed: the $p = 1.2 \text{ kg/m}^3$ air-density basis stated explicitly in §2; the CO₂ generation rate in §5 labelled *crew total* to foreclose a 4× misreading; the §5 marginal-ventilation costs labelled ERV'd vs bare-fresh, which are on different bases and were previously quoted side by side without distinction.
- **v1.3** — Terminology equivalence note added (§3): *dew-point indirect evaporative stage* / *dew-point IEC* recorded as equivalent to the *Maisotsenko cycle* / *M-cycle*, both forms retained so the disclosure depends on neither an eponym nor a trademark. Prime-mover reference in the X11 heat-grade ladder (§5) genericised to waste-heat platforms. No figure changed.
- **v1.4** — §9 expanded into a full conventions section: the identifier scheme (F/X/P, tests A–L, T/M, CHK, register row prefixes) with the never-renumbered and deliberate-gap rules stated explicitly, a symbol table, and an abbreviation key. A pointer to it is added above §1. Nothing is renamed and no figure changed — the notation was already in use throughout the lineage, but nowhere defined in one place.
- **v1.5** — Two figures added to the document every reader is told to open first, which until now carried none: §3 gains the dew-point-floor argument and the X2 regeneration split; §4 gains the dependency graph showing why the six balance equations must be solved simultaneously rather than as a hand chain. Both restate existing text — **no new number, and no figure changed**.